



histopathology cell injury-I

3rd class Lecture 2

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Objectives

By the end of this lecture, students will be able to

1. Analyze patterns of reversible and irreversible cell injury, identifying key features such as cellular swelling and nuclear changes.
2. Evaluate multiple causes of cell injury—including hypoxia, toxins, and immune reactions—by comparing their pathophysiological mechanisms and clinical impact.
3. Construct a complete summary table distinguishing between types of necrosis (e.g., coagulative, liquefactive, caseous), including morphology, causes, and affected tissues.
4. Accurately recall the stepwise events leading from cell injury to cell death and list at least 4 specific morphological features for each stage.
5. Describe the mechanisms and morphological features of apoptosis and explain how it differs from necrosis.
6. Identify histological evidence of apoptosis, fatty change, and calcification in tissue sections using specific visual and cellular criteria.

CELL INJURY

Definition and types of cell injury

- Cells actively interact with their environment, adjusting their structure and function to accommodate changing demands and extracellular stresses.
- The **intracellular environment** of cells is tightly regulated such that it remains fairly constant, a state referred to as **homeostasis**.
- As cells encounter **physiologic stresses** (increased workload in the heart) or potentially injurious conditions (nutrient deprivation), they can undergo **adaptation**, achieving a **new steady state** and preserving viability and function.
- If the adaptive capability is **exceeded** or if the external stress is inherently **harmful** or excessive, **cell injury** develops.

- Within certain limits, injury is **reversible**, and cells **return** to their stable baseline; however, if the stress is severe, persistent, or rapid in onset, it results in **irreversible injury** and death of the affected cells.
- **Cell death** is one of the most crucial events in the evolution of disease in any tissue or organ.
 - Cell death results from diverse causes, including ischemia (lack of blood flow), infections, toxins, and immune reactions.
- Cell death also is a normal and essential process in embryogenesis, the development of organs, and the maintenance of tissue homeostasis.

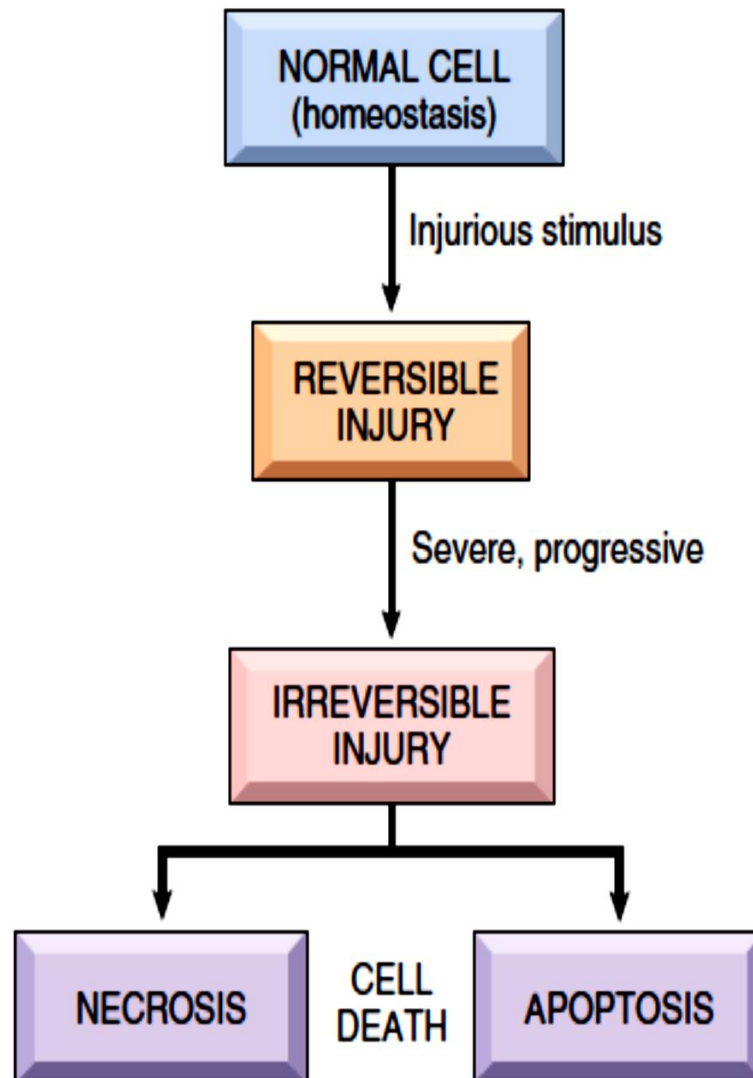


Fig. 2.2 Sequence of reversible cell injury and cell death. Necrosis and apoptosis are the two major pathways of cell death and are discussed in detail later.

CAUSES OF CELL INJURY

1. Hypoxia and ischemia:

- Hypoxia, which refers to **oxygen deficiency**, and
- ischemia, which means **reduced blood supply**.
- Both results in
 - deficiency of essential nutrients.
 - a build up of toxic metabolites.
- The most common cause of hypoxia is **ischemia** resulting from an arterial obstruction.
- oxygen deficiency can result from:
 - inadequate oxygenation of the blood, as in a variety of diseases affecting the lung, or
 - from reduction in the oxygen-carrying capacity of the blood, as with anemia of any cause, and carbon monoxide (CO) poisoning.

2. Toxins:

- Air pollutants, insecticides, CO, asbestos, cigarette smoke, ethanol, and drugs.
- Many drugs in therapeutic doses can cause cell or tissue injury in a susceptible patient or in many individuals if used excessively or inappropriately.
- Even glucose, salt, water and oxygen, can be toxic.

3. Infectious agents: All types of disease-causing pathogens, including viruses, bacteria, fungi, and protozoans, injure cells.

4. Immunologic reactions: Examples are

- **Autoimmune reactions** against one's own tissues,
 - **Allergic reactions** against environmental substances, and
 - **Excessive** or **chronic immune responses** to microbes.
- In all of these situations, immune responses elicit inflammatory reactions, which are often the cause of damage to cells and tissues.

5. Genetic abnormalities:

- the congenital malformations associated with **Down syndrome**.
- the single amino acid substitution in hemoglobin giving rise to **sickle cell anemia**.
- Deficiency of functional proteins, such as enzymes in **inborn errors of metabolism**.
- Accumulation of damaged DNA or misfolded proteins, both of which trigger cell death when they are beyond repair.

6. Nutritional imbalances.

- Protein–calorie insufficiency remains a major cause of cell injury.
- specific vitamin deficiencies are common even in developed countries with high standards of living.
- Excessive dietary intake may result in **obesity** and also is an important underlying factor in many diseases, such as **type 2 diabetes** mellitus and **atherosclerosis**.

7. Physical agents.

- Trauma, extremes of temperature, radiation, electric shock, and sudden changes in atmospheric pressure all have wide-ranging effects on cells.

8. Aging:

- Cellular senescence results in a diminished ability of cells to respond to stress and, eventually, the death of cells and of the organism.

SEQUENCE OF EVENTS IN CELL INJURY AND CELL DEATH

I. Reversible Cell Injury

- Reversible injury is the stage of cell injury at which the **deranged function** and **morphology** of the injured cells can **return to normal** if the damaging stimulus is removed.
- In reversible injury, cells and intracellular organelles typically become **swollen**
 - because they take in water as a result of the ***failure of energy-dependent ion pumps in the plasma membrane***, leading to
 - an **inability** to maintain ionic and fluid homeostasis.
- In some forms of injury, degenerated **organelles** and **lipids** may accumulate inside the injured cells.

Morphology of reversible cell injury

- The two main morphologic correlates of reversible cell injury are **cellular swelling** and **fatty change**.

1. Cellular swelling

- is commonly seen in cell injury associated with increased permeability of the plasma membrane.
- It may be difficult to appreciate with the light microscope, but it is often apparent at the level of the whole organ.
- When it affects many cells in an organ, it causes **pallor** (as a result of compression of capillaries), **increased turgor**, and an increase in organ weight.
- Microscopic examination:
 - **small, clear vacuoles** within the cytoplasm; these represent distended and pinched-off segments of the endoplasmic reticulum (**ER**).
 - The cytoplasm may become (**eosinophilic**), a change that becomes much more pronounced with progression to necrosis.
- This pattern of **nonlethal injury** is sometimes called **hydropic change** or **vacuolar degeneration**.

2. Fatty change:

- is manifested by the appearance of triglyceride containing lipid vacuoles in the cytoplasm.
- It is principally encountered in organs that are involved in lipid metabolism, such as the **liver**.

- **Other intracellular changes associated with cell injury include:**

- 1. plasma membrane** alterations such as

- blebbing,
- blunting, or
- distortion of microvilli, and
- loosening of intercellular attachments;

- 2. Mitochondrial** changes:

- swelling and
- the appearance of phospholipid-rich amorphous densities;

- 3. ER:**

- Dilation of the ER
- detachment of ribosomes and
- dissociation of polysomes

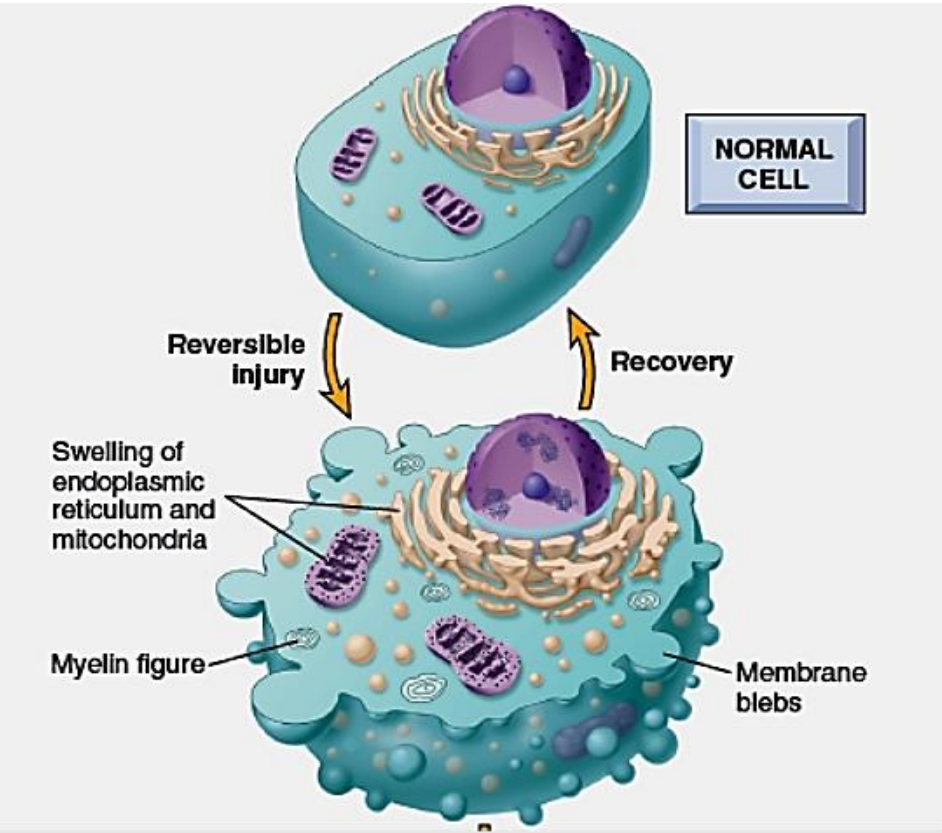
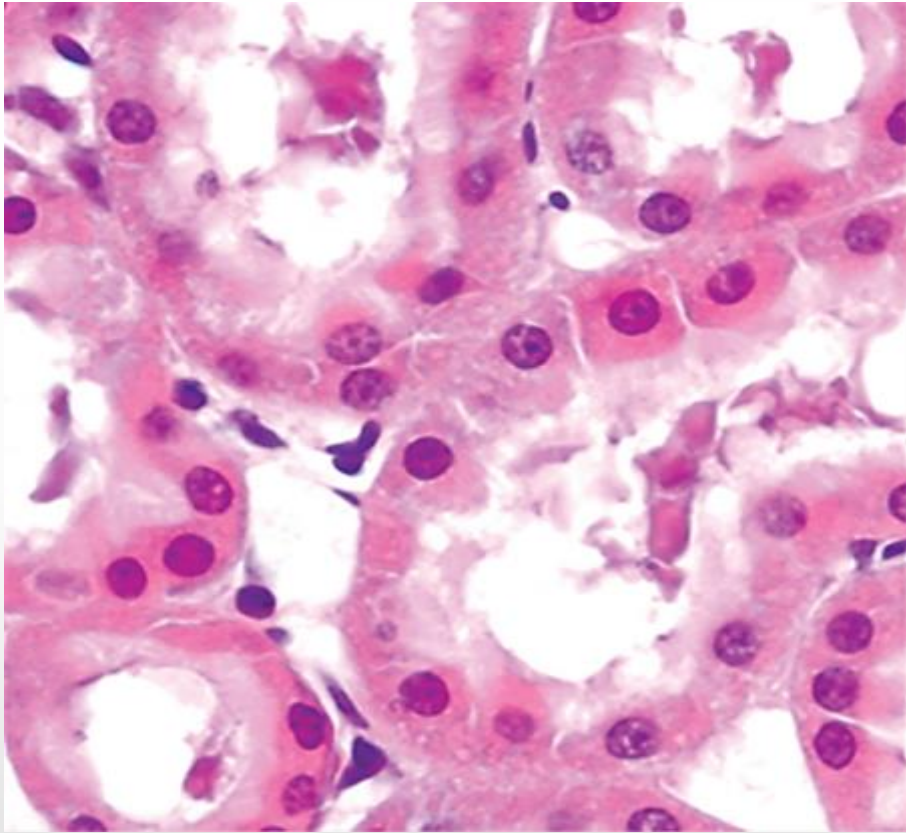
- 4. Nuclear alterations:**

- clumping of chromatin.

- 5. The cytoplasm:**

- contain so-called “myelin figures,” which are derived from damaged cellular membranes.
- The cytoplasm may become (eosinophilic), a change that becomes much more pronounced with progression to necrosis.

Reversible Cell Injury



Early (reversible) ischemic injury

- surface blebs,
- increased eosinophilia of cytoplasm
- swelling of occasional cells

Point of no return

- With persistent or excessive noxious exposures, injured cells pass a “**point of no return**” and undergo cell death.

II. The irreversibility of cell injury is characterized by three phenomena:

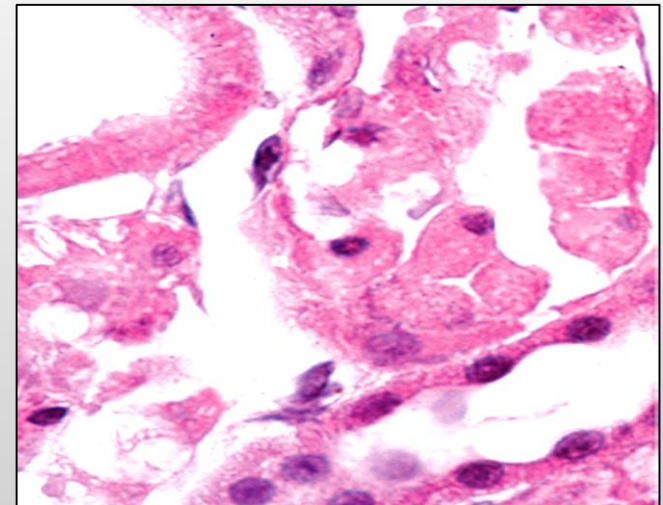
1. the *inability to restore mitochondrial function* (oxidative phosphorylation and [ATP] generation) even after resolution of the original injury;
2. the loss of structure and functions of the plasma membrane and intracellular membranes; and the loss of DNA and chromatin structural integrity.
3. Injury to lysosomal membranes results in the enzymatic dissolution of the injured cell, which is the culmination of necrosis.

Cell Death

- When cells are injured they die by different mechanisms, depending on the nature and severity of the insult.
- **A. Severe disturbances**, such as loss of oxygen and nutrient supply and the actions of toxins, infection and trauma cause a **rapid** and **uncontrollable** form of death that has been called “**accidental**” cell death.
 - The morphological manifestation of accidental cell death is **necrosis** (Greek, necros = death).
 - Necrosis is considered the **end result** of severe damage.

Necrotic (irreversible) injury of epithelial cells.

- loss of nuclei
- fragmentation of cells
- leakage of contents



B. In contrast,

- when the **injury is less severe**, or **cells need to be eliminated** during normal processes, they activate a set of molecular pathways that culminate in death.
- this is said to be “regulated” cell death.
- The morphologic appearance of most types of regulated cell death is **apoptosis**.
- In some instances, regulated cell death shows features of **both** necrosis and apoptosis, and has been called **necroptosis**.

APOPTOSIS

Know the signs.



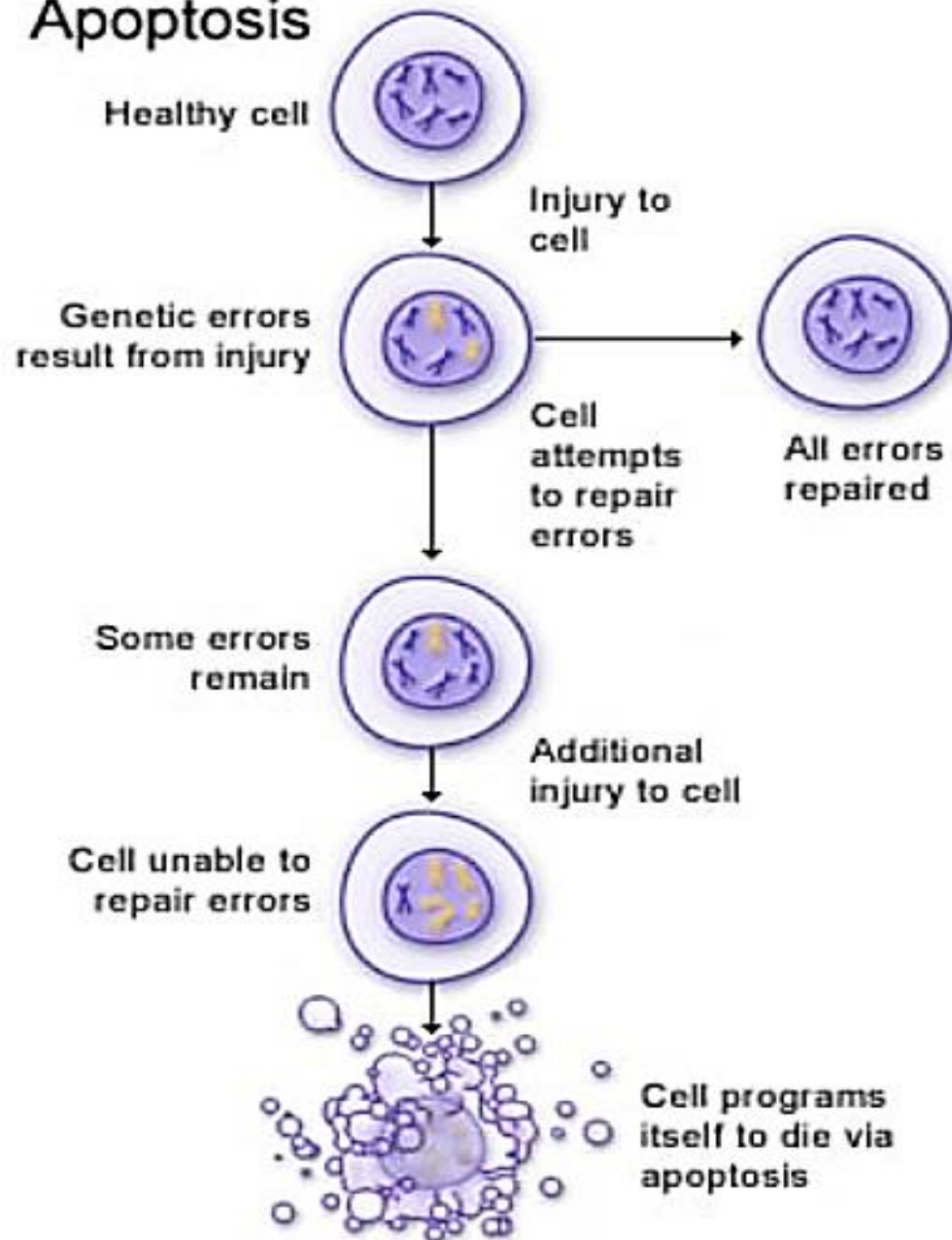
Apoptosis:

- is a process that **eliminates cells** with a variety of intrinsic abnormalities and promotes clearance of the fragments of the dead cells **without** eliciting an inflammatory reaction.
- This “**clean**” form of cell suicide occurs in **pathologic situations**:
 1. a cell’s DNA or proteins are damaged beyond repair or
 2. the cell is deprived of necessary survival signals.
- But unlike necrosis, which is always an indication of a pathologic process, apoptosis also occurs in **healthy tissues**.
- It serves to:
 1. eliminate unwanted cells during normal development and
 2. to maintain constant cell numbers, so it is not necessarily associated with pathologic cell injury.
- These types of physiologic cell death are also called **programmed cell death**.

APOPTOSIS

- This form of cell death is a regulated suicide program.
- In apoptosis the relevant cells **activate enzymes** capable of degrading the cells' own nuclear DNA and other nuclear and cytoplasmic proteins.
- The plasma membrane of the apoptotic cell remains **intact**, but is altered in such a way that the **cell** becomes a **target** for **phagocytes**.
- The dead cell is **rapidly cleared** **before** its contents have leaked out.
- cell death by this pathway does not elicit an inflammatory reaction in the host.

Apoptosis

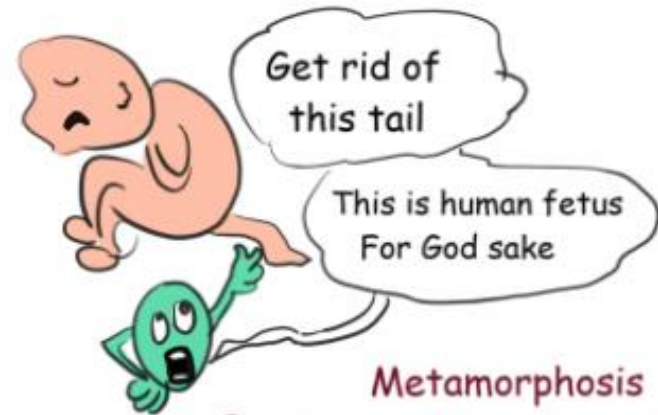




Lymphocyte



Neutrophil



Metamorphosis

1. After Serving the purpose

2. Self destructive cells

Physiological Causes of Apoptosis

Creative-Med-Doses ©2019 Priyanga Singh

3.

Embryogenesis

Odd cells

4.

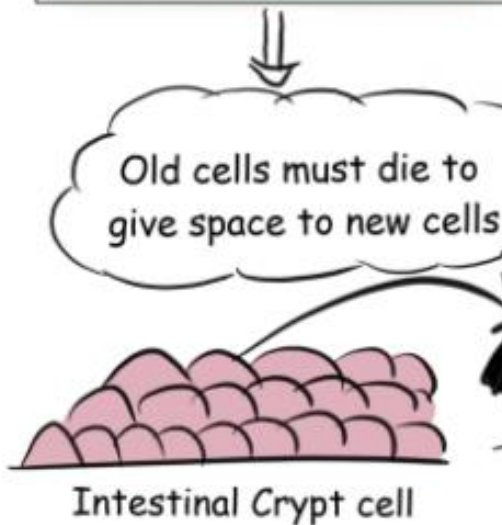




Castration

Creative-Med-Doses
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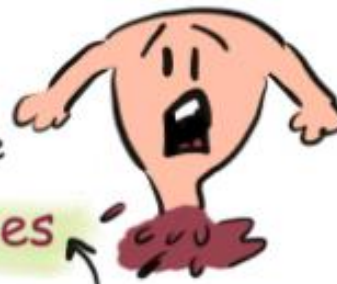
5. Old cells in highly Proliferating organs



Old cells must die to give space to new cells

Low estrogen
Low progesterone
Endometrial shedding

Menses



Ovarian atresia



Low estrogen

6. Hormonal

Menopause

Physiological Causes of Apoptosis

Breasts after weaning

No respect for elders
Huh, new generation...



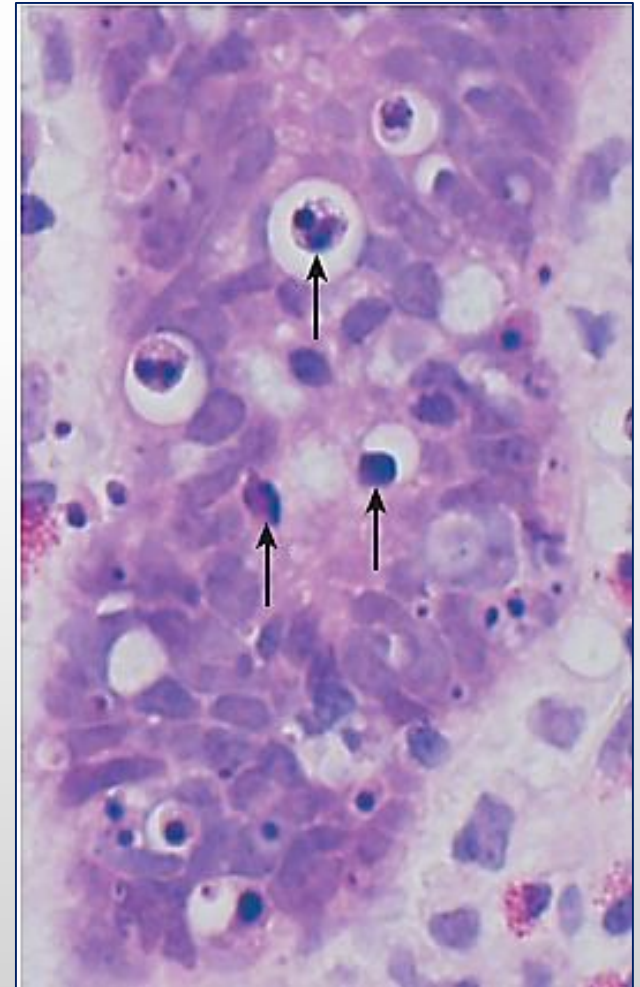
Less Oxytocin

Morphology of apoptotic cell

- In H&E-stained tissue sections, the nuclei of apoptotic cells show various stages of chromatin condensation and aggregation and, ultimately, karyorrhexis.
- The cells
 1. Rapidly **shrink**.
 2. Form **cytoplasmic buds**.
 3. Fragment into **apoptotic bodies**
- Apoptotic bodies are composed of membrane bound pieces of cytosol and organelles.

Morphologic appearance of apoptotic cells.

- Apoptotic cells (some indicated by *arrows*) in colonic epithelium.
- the fragmented nuclei with condensed chromatin and the shrunken cell bodies, some with pieces falling off.



Causes of Apoptosis

I. Apoptosis in Physiologic Situations.

II. Apoptosis in Pathologic Conditions.

Condition	Mechanism of Apoptosis
Physiologic	
During embryogenesis	Loss of growth factor signaling (presumed mechanism)
Turnover of proliferative tissues (e.g., intestinal epithelium, lymphocytes in bone marrow, and thymus)	Loss of growth factor signaling (presumed mechanism)
Involution of hormone-dependent tissues (e.g., endometrium)	Decreased hormone levels lead to reduced survival signals
Decline of leukocyte numbers at the end of immune and inflammatory responses	Loss of survival signals as stimulus for leukocyte activation is eliminated
Elimination of potentially harmful self-reactive lymphocytes	Strong recognition of self antigens induces apoptosis by both the mitochondrial and death receptor pathways
Pathologic	
DNA damage	Activation of proapoptotic proteins by BH3-only sensors
Accumulation of misfolded proteins	Activation of proapoptotic proteins by BH3-only sensors, possibly direct activation of caspases
Infections, especially certain viral infections	Activation of the mitochondrial pathway by viral proteins Killing of infected cells by cytotoxic T lymphocytes, which activate caspases

Features of Necrosis and Apoptosis

Feature	Necrosis	Apoptosis
Cell size	Enlarged (swelling)	Reduced (shrinkage)
Nucleus	Pyknosis → karyorrhexis → karyolysis	Fragmentation into nucleosome-sized fragments
Plasma membrane	Disrupted	Intact; altered structure, especially orientation of lipids
Cellular contents	Enzymatic digestion; may leak out of cell	Intact; may be released in apoptotic bodies
Adjacent inflammation	Frequent	No
Physiologic or pathologic role	Invariably pathologic (culmination of irreversible cell injury)	Often physiologic means of eliminating unwanted cells; may be pathologic after some forms of cell injury, especially DNA and protein damage

Mechanisms of Apoptosis

- Apoptosis is regulated by biochemical pathways that control:
 1. the **balance** of death- and survival-inducing signals.
 2. the **activation** of enzymes called **caspases**.
- Two pathways converge on caspase activation:
 1. The Mitochondrial pathway.
 2. The Death receptor pathway.
- The end result of apoptotic cell death is the **clearance of apoptotic bodies** by **phagocytes**.

Cell Death

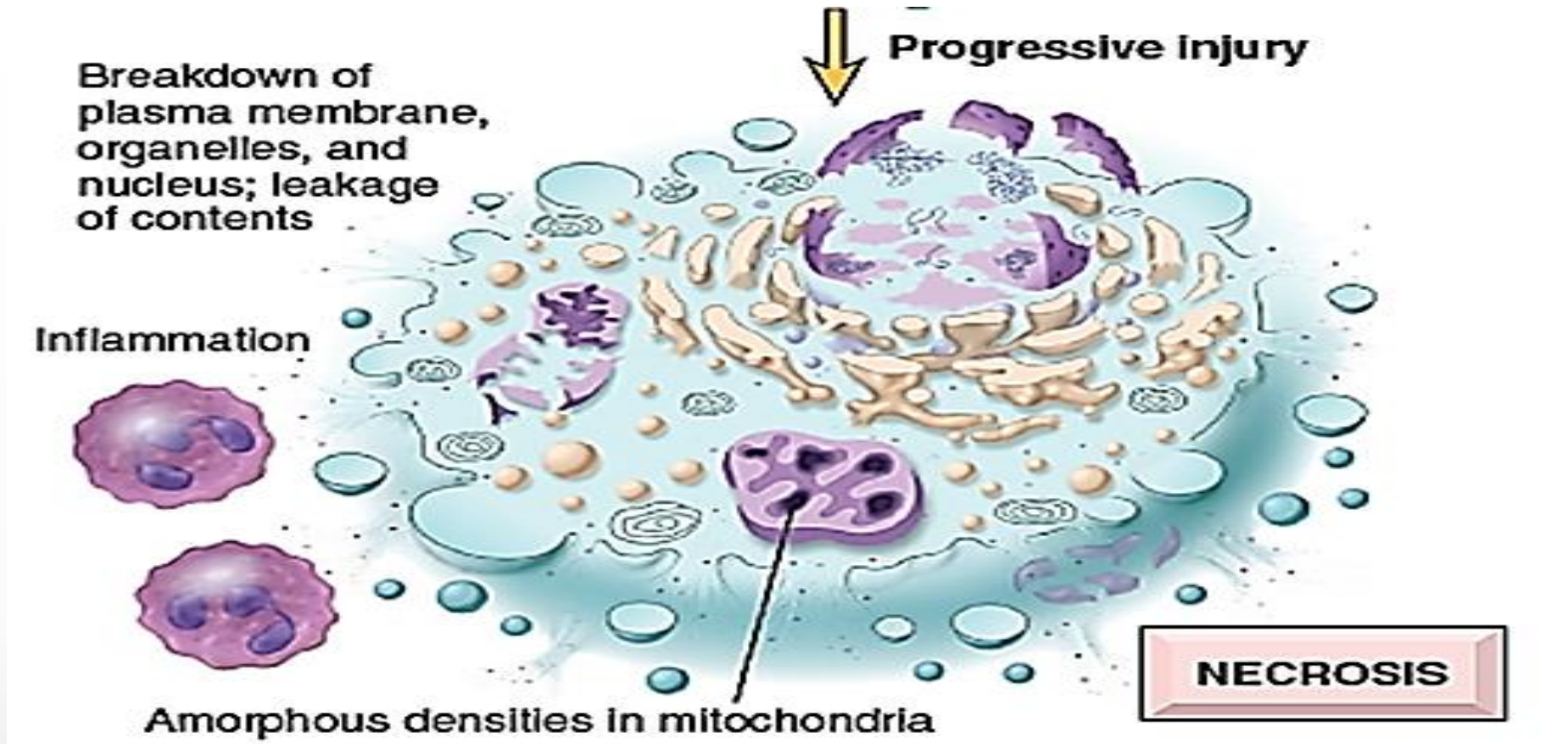
- The **cellular function** maybe **lost** long before cell death occurs, so
- the morphologic changes of cell injury (or death) **lag far behind loss of function** and viability.

For example, **myocardial cells**:

- become **non-contractile** after 1 - 2 minutes of **ischemia**, but
- may not **die** until 20 to 30 minutes of ischemia.
- E.M: **Morphologic features** indicative of the death of ischemic myocytes appear within 2 to 3 hours after the death of the cells.
- are evident by light microscopy until 6 to 12 hours later.

Necrosis

- Necrosis is a form of cell death in which cellular membranes fall apart, and cellular enzymes leak out and ultimately digest the cell.
- Necrosis elicits a local host reaction, called *inflammation*.
 - Inflammation is induced by **substances** released from dead cells.
 - Inflammation serves to eliminate the debris and start the repair process.
- The enzymes responsible for digestion of the cell are derived from:
 1. **lysosomes** and may come from the
 2. **Dying cells** themselves or from
 3. **Leukocytes** recruited as part of the inflammatory reaction.
- Necrosis often is the culmination of reversible cell injury that cannot be corrected.



The mechanisms of necrosis include:

1. Failure of energy generation in the form of ATP because of reduced oxygen supply or mitochondrial damage;
2. Damage to cellular membranes, including the plasma membrane and lysosomal membranes, which results in leakage of cellular contents including enzymes.
3. Irreversible damage to cellular lipids, proteins, and nucleic acids, which may be caused by reactive oxygen species (ROS); and others.

Morphologic features of necrosis

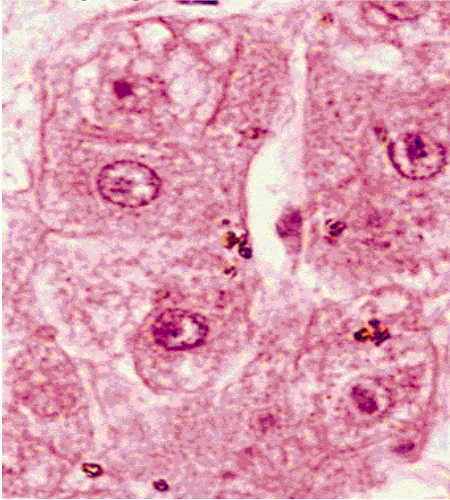
I. Cytoplasmic changes:

- the necrotic cells show **increased** cytoplasmic **eosinophilia** (i.e. it appears deep pink in color than normal cells).
 - This is attributable in part:
 - to increased binding of eosin to denatured cytoplasmic proteins and
 - in part to loss of the basophilia that is normally imparted by the RNA in the cytoplasm (basophilia is the blue staining from the hematoxylin dye).
- The cell may have a **more homogeneous** appearance than viable cells, mostly because of the **loss of glycogen** particles.

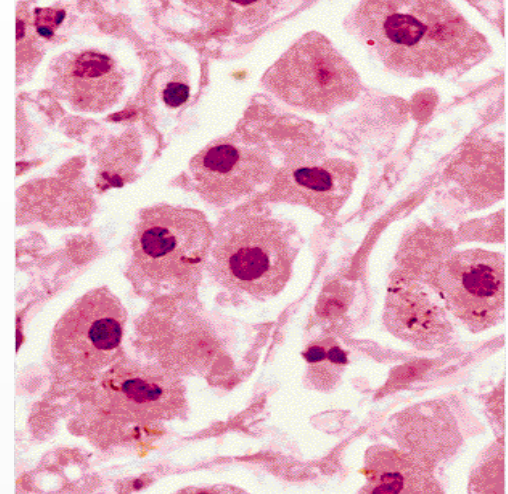
II. Nuclear changes :

- these assume one of three patterns, all due to **breakdown of DNA** and **chromatin**.
 - **Pyknosis**: characterized by nuclear shrinkage and increased basophilia; the DNA condenses into a solid shrunken mass.
 - **Karyorrhexis**: the pyknotic nucleus undergoes fragmentation.
 - **Karyolysis**: the basophilia of the chromatin become pale secondary to deoxyribonuclease (DNase) activity.

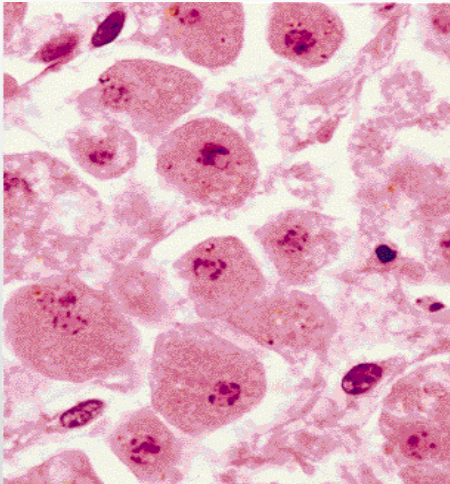
Liver cell necrosis: Nuclear changes



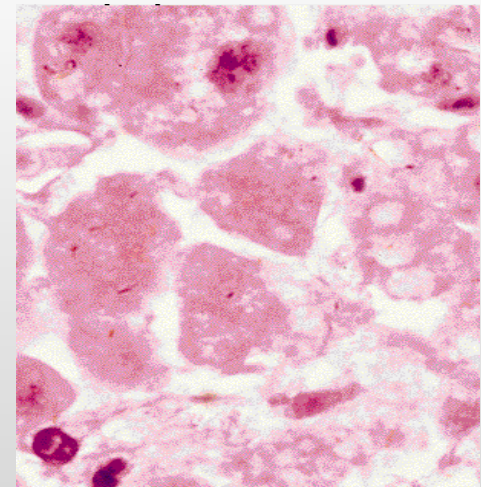
normal



pyknosis



karyorrhexis

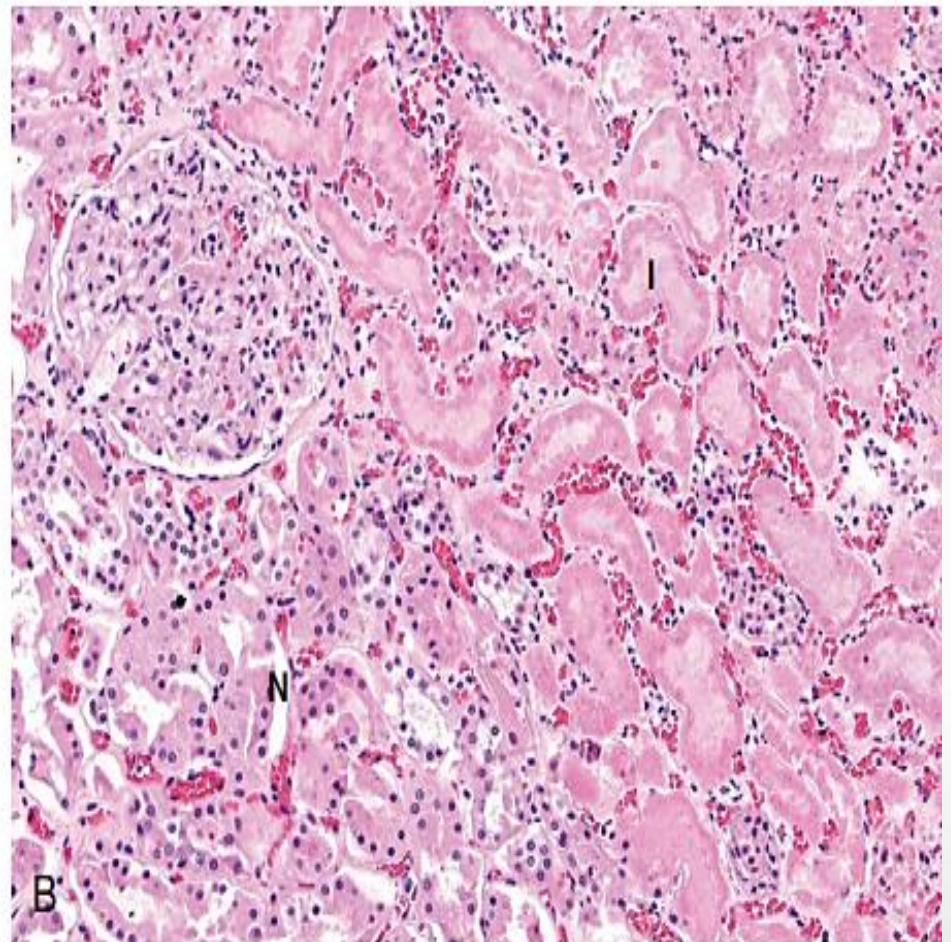


karyolysis

Patterns of Tissue Necrosis

1. Coagulative necrosis is characterized

- **Grossly:** by **firmness** of the affected tissue and
- **Microscopically:**
 - by loss of the cellular fine structural details but **preservation** of the basic **tissue architecture**.
 - The necrotic cells show **homogeneously eosinophilic cytoplasm** and are **devoid of nuclei**.
 - Coagulative necrosis is characteristic of ischemic damage in **all solid** organs **except the brain**.



Coagulative necrosis.

(A) A wedge-shaped kidney infarct (yellow) with preservation of the outlines.

(B) Microscopic view of the edge of the infarct, with normal kidney (N) and necrotic cells in the infarct (I). The necrotic cells show preserved outlines with loss of nuclei, and an inflammatory infiltrate is present (difficult to discern at this magnification)

2. **Liquifactive necrosis:**

- is characterized by **complete digestion** of the dead cells, resulting in transformation of the affected tissue into thick liquid.
- It is enclosed within a **cystic cavity**.



This is an example of liquefactive necrosis; the affected area is wedge-shaped, pale, soft & cystic.

- This type of necrosis is seen in two situations
 - a. **Focal pyogenic bacterial infections**; the liquefied material is frequently creamy yellow and is called pus.
 - b. **Ischemic destruction of the brain tissue**.

3. Gangrenous necrosis (gangrene):

- is not a distinctive pattern of cell death;
- It is usually applied to **a limb**,
- A leg that has lost its blood supply and has undergone coagulative necrosis involving multiple tissue layers (**dry gangrene**).
- When bacterial infection is superimposed, coagulative necrosis is modified by the liquefactive action of the bacteria (**wet gangrene**).

4. Caseous necrosis:

- The tissue architecture is **completely lost** and cellular outlines **cannot** be detected.
- It is encountered in foci of tuberculous infection.
- The term "caseous" (cheese-like) is derived from the friable yellow-white appearance of the area of necrosis.
- Microscopically, the necrotic focus appears as **pinkish**, and **granular** in appearance.
- Caseous necrosis is often bordered by an inflammation.

Ganagrene of lower limb



Dry gangrene



Wet gangrene

Caseous necrosis



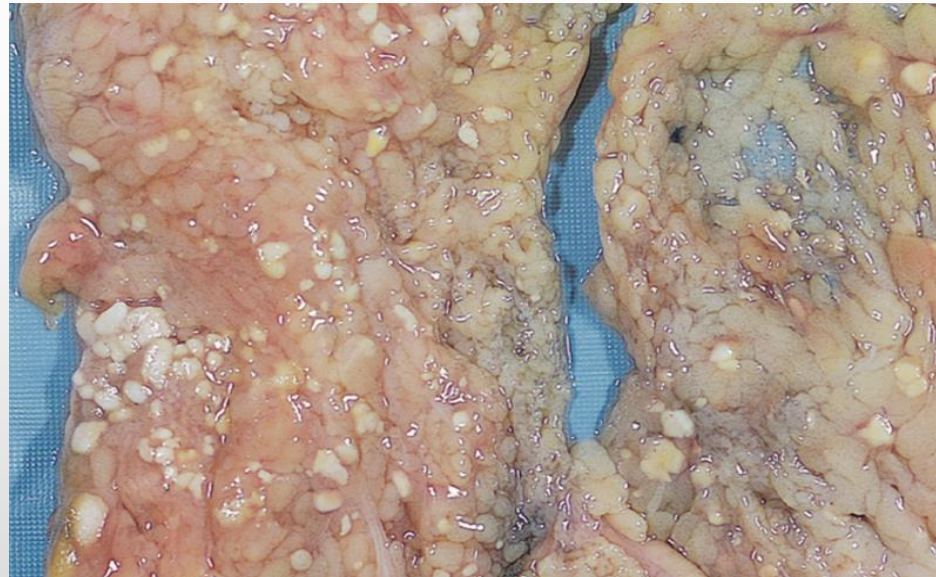
A tuberculous lung with a large area of caseous necrosis containing yellow-white and cheesy debris.

5. Fat necrosis: is typically seen in

a. acute pancreatitis and results from release of activated pancreatic lipases into the pancreas and the peritoneal cavity that liquefy the membranes of fat cells in the pancreas.

- Grossly: produce grossly visible chalky white areas.
- Microscopically, the foci of necrosis contain vague outlines of necrotic fat cells with bluish calcium deposits.

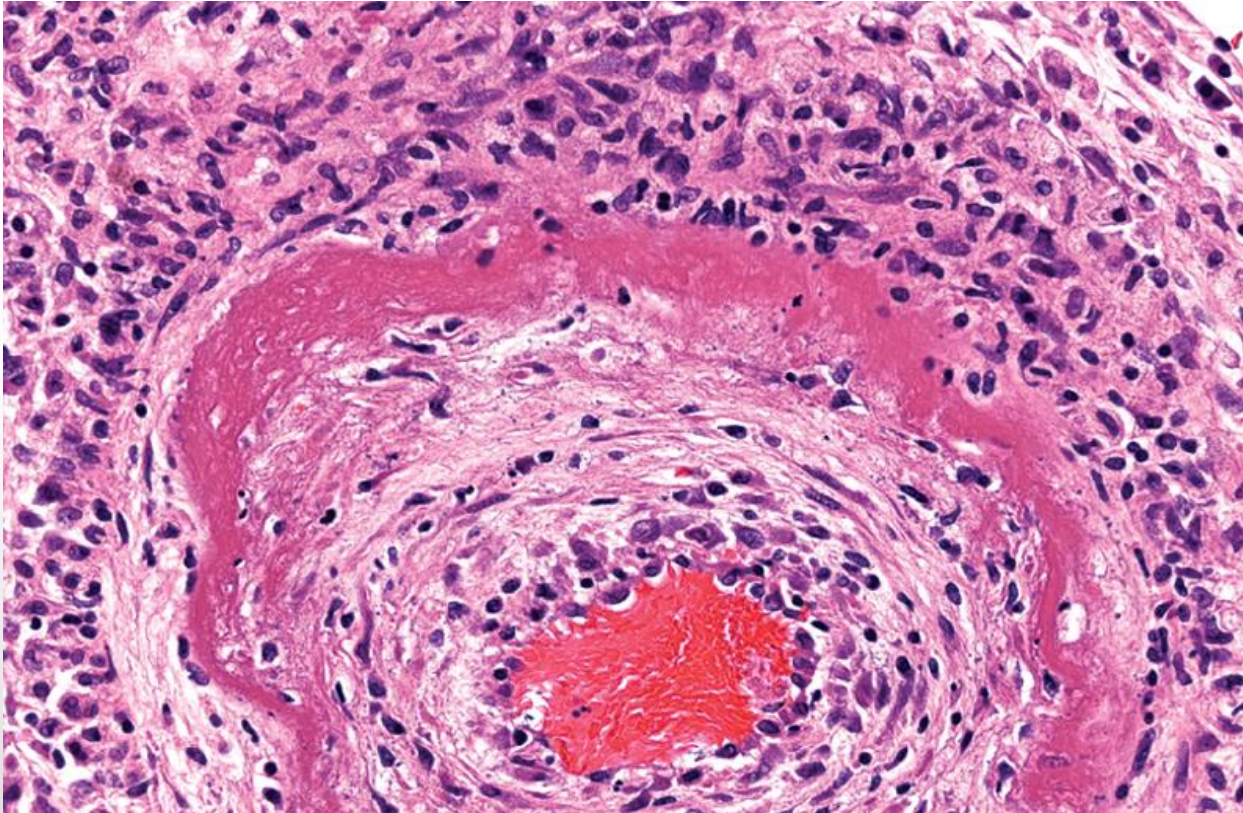
b. fat necrosis is seen in female breasts; preceded by a history of trauma (traumatic fat necrosis).



The areas of white chalky deposits represent foci of fat necrosis with calcium soap formation (saponification) at sites of lipid breakdown in the mesentery.

6. Fibrinoid necrosis:

- is typically seen in **immune reactions** involving **blood vessels**.
- Deposits of immune complexes, together with fibrin result in a homogeneous bright pink appearance.



The wall of the artery shows a circumferential bright pink area of necrosis with protein deposition and inflammation (dark nuclei of neutrophils).

